

AAA -- Authorization -- Guided Notes

Unit tp-6.6 | CompTIA Network+ N10-009 Obj. FC0-U71 6.1 | N10-008 FC0-U71 6.1

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

Authorization (the second A in AAA) determines what a verified user is _____ to do after they authenticate. Authentication proves _____ the user is; authorization defines _____ they can access and what _____ they can perform.

Question 2

Role-Based Access Control (RBAC) assigns permissions based on _____ rather than individual identity. A nurse role has different access than a _____ role. Permissions are granted to _____, and users inherit permissions by being assigned a role. This is easier to _____ at scale than per-user permissions.

Question 3

The principle of _____ privilege states that each user or system component should have only the _____ permissions required to perform their job -- nothing extra. This limits _____ in the event of a compromised account. If an accountant is compromised, they should not be able to access _____ source code or patient records.

Question 4

Mandatory Access Control (MAC) uses _____ labels (e.g., Top Secret, Confidential) assigned by the system or administrator. Users can only access resources at or _____ their clearance level. MAC is common in _____ and government environments. Discretionary Access Control (DAC) lets the _____ owner set permissions.

Question 5

Privilege _____ is a common attack where an attacker exploits a vulnerability to gain _____ access than authorized. Horizontal escalation means gaining access to _____ user accounts at the same level. Vertical escalation means gaining _____ (admin or root) privileges from a standard user account.

Question 6

An Access Control List (ACL) is a list attached to a resource (file, folder, network port) that specifies which _____ or _____ are allowed or denied access. On Linux file systems, each file has _____ (read/write/execute) permissions for owner, _____, and others. On Windows, NTFS uses ACLs to control file and folder access.

Question 7

Separation of duties requires that critical tasks are _____ across multiple people so no single person can complete a sensitive action alone. For example, the person who _____ a payment should not be the same person who _____ it. This is a key internal control that reduces _____ and financial fraud.

Question 8 -- Real World Application

REAL WORLD: A software company gives all engineers admin rights to the production database because it is more convenient during development. One engineer's laptop is infected with malware. The malware uses the engineer's stored database credentials to export the entire customer database containing 2 million email addresses and passwords. Explain the authorization failure and describe two specific controls that would have limited the impact.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/tp-6-6-aaa---authorization/t  
p-6-6-aaa---authorization-answer-key.pdf
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